Sadiq S.

Practical 5

Aim: To develop an IoT application that will raise an alarm whenever with going to raining outside and also show the amount of rainfall

Theory:

Water Sensor: Water sensor is designed for water detection, which can be widely used in sensing rainfall, water level, and even liquid leakage.

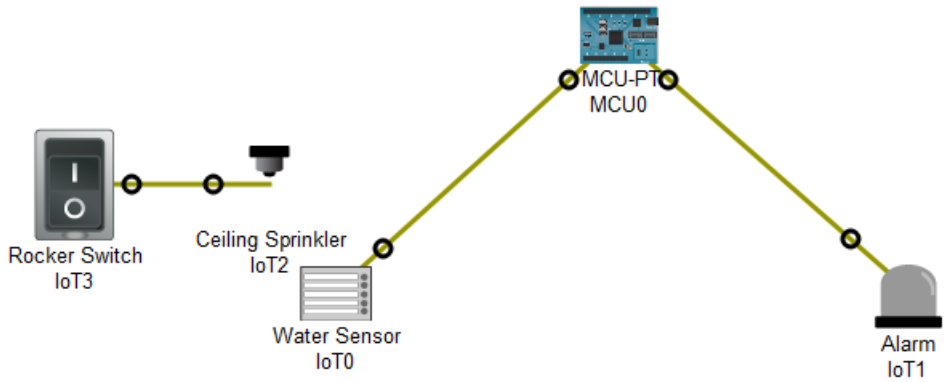
Rocker switches: Rocker switches are commonly used to start or stop and On or Off the flow of current in the circuit with every push on either side. Rocker switches are often contradicted with a general toggle or trip switches.

In this practical we will raise an alarm if there is raining outside. Also we will calculate the amount of rainfall.

Requirements:

- 1 Water Sensor
- 1 Microcontroller Unit
- 1 Ceiling Sprinkler
- 1 Rocker Switch
- 1 Alarm

Diagram:



Coding:

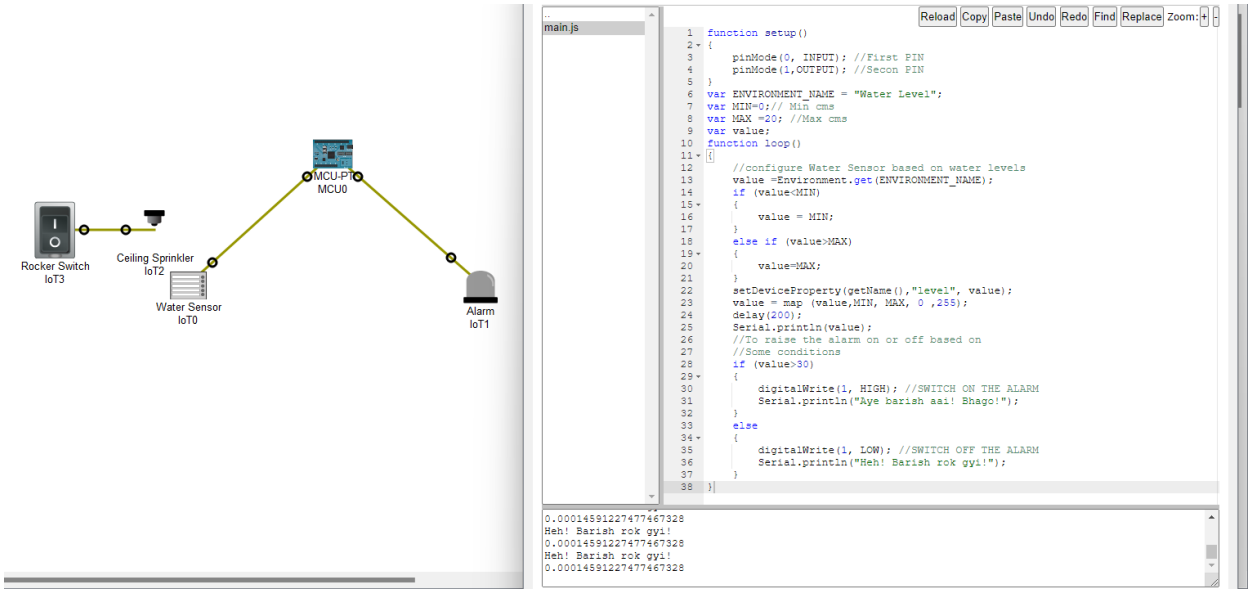
To write the code we have click on MicroController Unit. Then go to Programming tab. Create a new JavaScript file and write the code.

```
function setup()
{
                    pinMode(0, INPUT); //First PIN
                    pinMode(1,OUTPUT); //Secon PIN
}
var ENVIRONMENT_NAME = "Water Level";
var MIN=0;// Min cms
var MAX =20; //Max cms
var value;
function loop()
{
                    //configure Water Sensor based on water levels
                    value =Environment.get(ENVIRONMENT_NAME);
                    if (value<MIN)
                    {
                        value = MIN;
                    }
                    else if (value>MAX)
                    {
                        value=MAX;
                    }
                    setDeviceProperty(getName(),"level", value);
                    value = map (value,MIN, MAX, 0 ,255);
                    delay(200);
                    Serial.println(value);
                    //To raise the alarm on or off based on
                    //Some conditions
                    if (value>30)
```

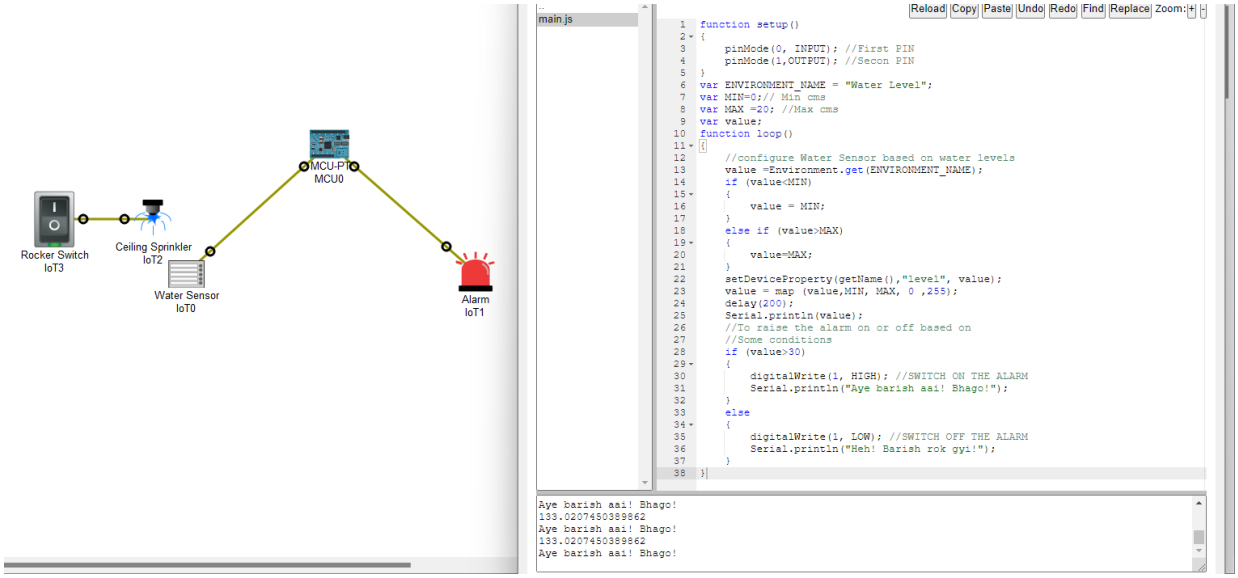
```
        {
            digitalWrite(1, HIGH); //SWITCH ON THE ALARM
            Serial.println("Aye barish aai! Bhago!");
        }
        else
        {
            digitalWrite(1, LOW); //SWITCH OFF THE ALARM
            Serial.println("Heh! Barish rok gyi!");
        }
}
```

Output:

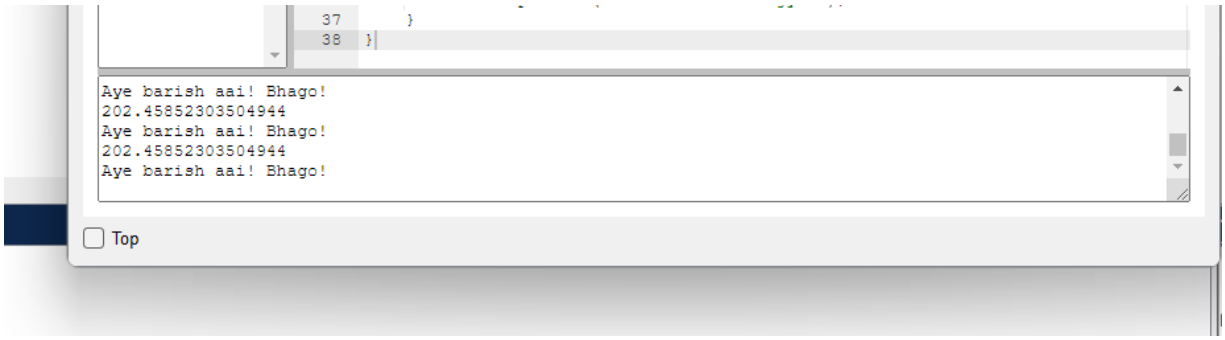
When the rocker switch is off then there is no raining. It will show some small amount.

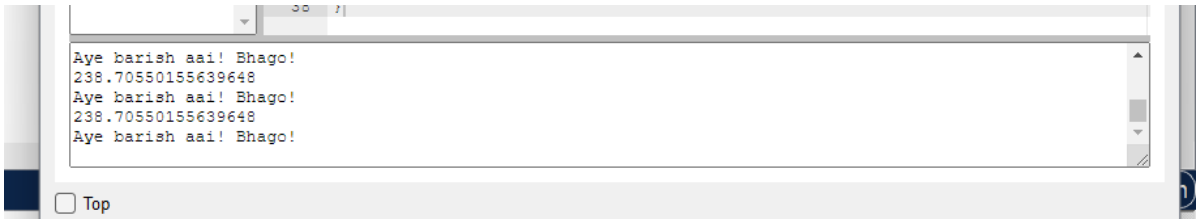


When the rocker switch is on then the rainfall will start outside.

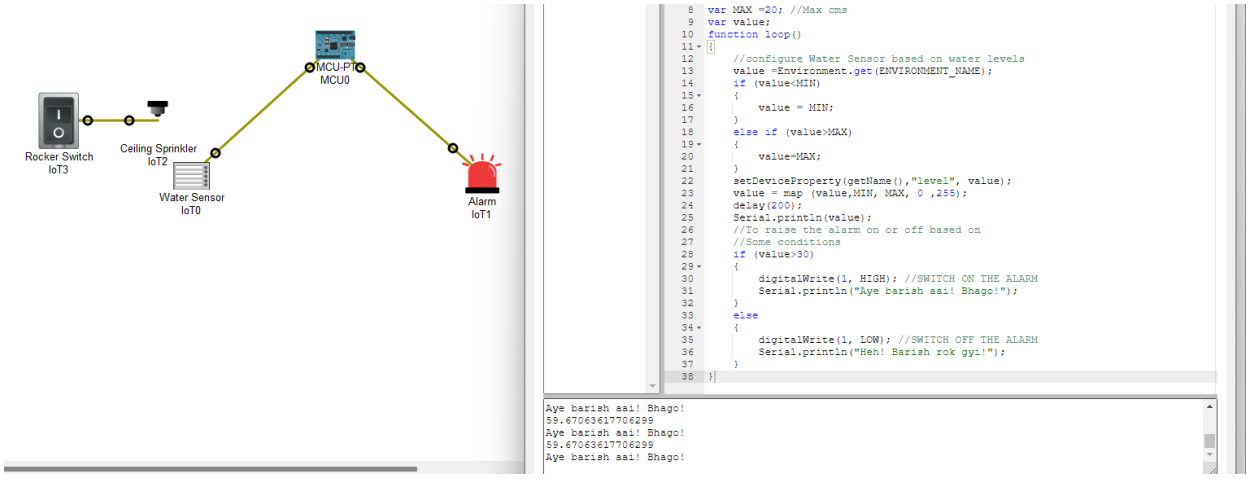


The amount of rainfall will keep on increasing as long as the rocker switch is on.

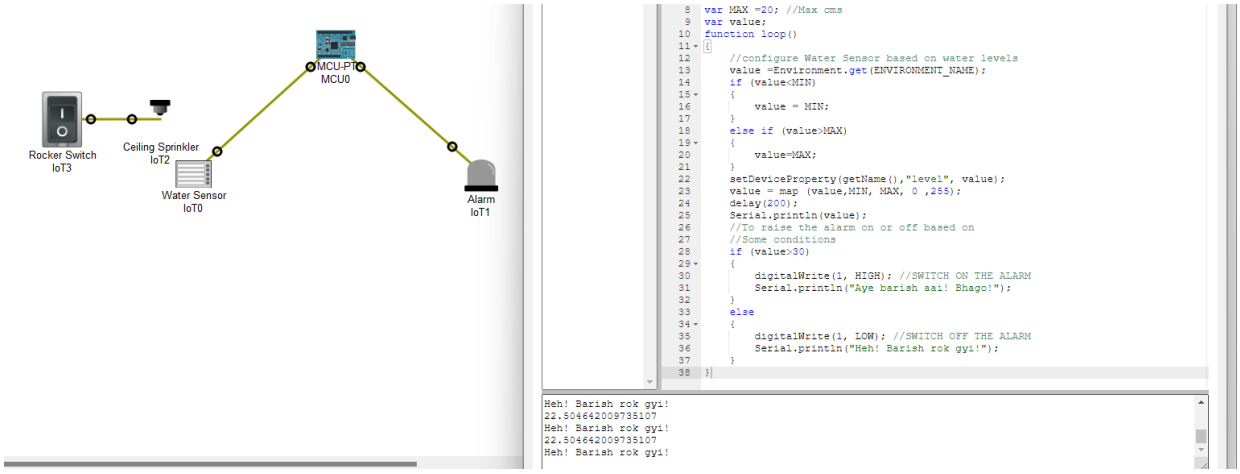




Once the rocker switch is turned off. The amount of rainfall will start to decrease. But the alarm will remain in on mode.



Once the amount of rainfall goes below 30. The alarm will turn off.



And the rainfall will keep on decreasing until it reaches 0.000145.

After reaching 0.000145 the rainfall won't decrease any further.